

Multi-Message Quantum Broadcast with Fine-Grained Access Control

Sayantan Ganguly¹, Shion Samadder Chaudhury²

Abstract—Several subscription-based systems require fine-grained, dynamic access control, which traditional broadcast encryption schemes fail to handle efficiently. Attribute-based and policy-based access structures offer a flexible and scalable solution by encoding access rules directly into the encryption mechanism. With the advent and development of quantum networks, attribute-based and policy-based quantum broadcast encryption (PBQBE) becomes necessary for enforcing such controls securely in a post-quantum world. In this paper, we present a general construction of quantum broadcast encryption for multiple messages simultaneously, based on routing and traversing a given attribute-based and, consequently, a policy-based access tree when a user satisfies a unique policy. Our scheme can handle dynamic systems with frequently changing access patterns and can be deployed seamlessly with classical systems. Compared to existing constructions, our scheme offers hierarchical access, policy control, scalability, and revocation while maintaining provable security guarantees.

Index Terms—Attribute-based encryption, Broadcast encryption, Policy access-tree traversal, Policy-based quantum broadcast encryption.

I. INTRODUCTION

THE wide use of subscription-based digital services has highlighted the need for fine-grained and dynamic access control mechanisms. While foundational for one-to-many secure communication, traditional broadcast encryption (BE) schemes often struggle with the efficiency and flexibility required by these modern systems, particularly when managing frequently changing user privileges and access rules. Policy-based access control, notably through Attribute-Based Encryption (ABE) and Policy-Based Broadcast Encryption (PBBE), offers a more scalable and expressive solution by directly embedding access policies into the cryptographic framework. This allows data to be encrypted such that only users whose attributes or credentials satisfy the specified policy can decrypt, enabling intricate control over information dissemination.

The advent of quantum computing poses a significant threat to the security of classical cryptographic systems, including existing BE and PBBE schemes, which are primarily built upon computational hardness assumptions like factoring or discrete logarithms. As quantum networks and quantum information processing capabilities progress, developing quantum-resistant and inherently quantum solutions for broadcast com-

munication becomes important. Policy-based quantum broadcast encryption (PBQBE) is thus essential for enforcing complex access controls securely in a post-quantum world, ensuring the confidentiality and integrity of sensitive information transmitted over emerging quantum channels. Addressing this need, quantum cryptographic primitives are being explored not only for their potential resistance to quantum attacks but also for their unique capabilities in information processing.

The work [1] by Broadbent et al. on quantum private broadcasting provided novel solutions for transmitting quantum states to multiple recipients with information-theoretic security, using unitary t -designs for efficient key management. Quantum Attribute-Based Encryption schemes have also been proposed, demonstrating the feasibility of policy-based access control in the quantum realm, though often with considerations regarding specific trust models, computational overhead, or the range of supported policies. Many existing approaches, both classical and quantum, still face challenges concerning dynamic user management, efficient revocation, support for simultaneous multi-message broadcasting under diverse policies, and seamless integration with classical infrastructures, while maintaining provable security and manageable implementation complexity.

In this paper, we introduce a policy-based quantum broadcast encryption designed to address these challenges. Specifically, we consider the following scenario.

- Suppose there are multiple quantum messages to broadcast, each associated with a distinct access policy. We consider a setting where a user or a group of users possesses attributes that satisfy a particular policy, and only the message corresponding to that policy should be decryptable by them. We construct a quantum broadcast encryption scheme based on access tree traversal, in which decryption is possible only when the user's attributes satisfy the specific policy linked to the target message.

Compared to existing classical and quantum constructions, our proposed schemes aim to deliver a compelling combination of hierarchical access control, expressive policy enforcement, enhanced scalability, and efficient revocation mechanisms. We provide a rigorous analysis of their security properties.

Related Works: The evolution of secure multi-user communication has spanned from classical to quantum domains, laying the groundwork for our proposed policy-based quantum broadcast encryption. Initially, Broadcast Encryption (BE), introduced by Fiat and Naor [2] and surveyed by Horwitz [3], provided one-to-many secure communication for static user sets. However, the demand for dynamic and fine-grained

¹Institute for Advancing Intelligence(IAI), TCG CREST, EP & GP Block, Sector - V, Salt Lake, Kolkata-700091, West Bengal, India.

¹Ramakrishna Mission Vivekananda Educational and Research Institute (RK-MVERI), Belur, West Bengal, India.

²Academy of Scientific and Innovative Research (AcSIR), Ghaziabad, India. Corresponding Author Email: ¹sayantan.ganguly.90@tcgcrest.org Corresponding Author Email: ²chaudhury.shion@gmail.com

access inspired the development of Attribute-Based Encryption (ABE). Sahai and Waters' Fuzzy Identity-Based Encryption [4] and Bethencourt, Sahai, and Waters' Ciphertext-Policy ABE (CP-ABE) [5] enabled policies to be embedded within the ciphertext or keys, allowing decryption only if a user's attributes satisfied the policy. Decentralized ABE schemes further emerged to address trust concerns [6]. More recently, classical BE and ABE schemes have seen advancements, including identity-based schemes for IoT (Internet of Things) [7] and post-quantum secure, recipient-revocable BE [8], [9], along with optimal lattice-based constructions for BE and CP-ABE [10].

Quantum Broadcast Encryption (QBE) schemes were developed for secure quantum information dissemination. Early contributions like Jian et al.'s work [11] discussed general quantum broadcast communication. Later research explored entanglement-assisted private communication [12], quantum broadcast schemes via cluster states [13], and those utilizing partially entangled channels [14]. Sukeno et al. [15] recently examined broadcasting entangled states for authentication and cryptography. Developments also include quantum blind signature schemes for broadcasting [16], [17] and quantum broadcast channels handling individualized and confidential messages [18].

An important work in quantum broadcasting is Quantum Private Broadcasting (QPB) by Broadbent et al. [1], [19], [20]. This scheme provides information-theoretic security by employing unitary t -designs [21], [22], ensuring that fewer than t colluding recipients gain no information about the broadcast state. Extending ABE's flexibility to the quantum domain, Quantum Attribute-Based Encryption (QABE) schemes have been proposed by Samadder Chaudhury et al.'s Quantum Ciphertext-Policy ABE [23]. The works on quantum walks [24], [25], serve as powerful tools for quantum information processing, including universal computation [26].

Our Contribution: In this work, we define a policy-based access tree, where each leaf node defines a unique policy, and each policy is defined by an attribute-based access tree. Based on the policy-based access tree, we construct a quantum broadcast encryption for single and multiple messages, traversing through a policy-based access tree using flexible routing to a policy-driven framework for dynamic access control in emerging quantum network environments that integrates policy-based access control with quantum information processing.

Note that our constructions are general and can be modified based on different scenarios and purposes.

Paper Organization: This paper is organized as follows: Section I introduces the core concepts, applications, and motivation for policy-based quantum broadcast encryption (PQBE), along with related work and our contributions. Section II provides essential notations and preliminaries. In Section III, we detail our multi-message broadcast encryption scheme by traversing a policy-based access tree. In section V, we analyze the quantum resource usage and complexity of our proposed constructions. Finally, in section VI we concluded our paper discussing the limitation of applicability of our construction

in the present quantum infrastructure and proposed future research problems in this direction.

II. PRELIMINARIES

A. Notations

Let $\mathcal{H}_V$ denote a Hilbert space of dimension $|V|$ with standard basis vectors $\{|v\rangle : v \in \{0, 1, \dots, |V| - 1\}\}$. The set of all density operators acting on this space is written as $\mathcal{D}(\mathcal{H}_V)$, while the set of linear operators is denoted by $\mathcal{L}(\mathcal{H}_V)$. For a composite system involving registers V and M , we write the total Hilbert space as $\mathcal{H}_V \otimes \mathcal{H}_M$, and use ρ_{VM} to represent a density operator on this space. The marginal state on V is obtained by taking the partial trace over M , i.e., $\rho_V = \text{Tr}_M(\rho_{VM})$. We denote $U(d)$ to denote the group of all $d \times d$ unitary matrices.

Quantum channels are maps that are completely positive and trace-preserving (CPTP). The distinguishability of quantum states is measured using the trace norm: $\|A\|_1 = \text{Tr}(\sqrt{AA^\dagger})$, which equals the sum of singular values of A . The operator norm, or infinity norm, is given by the largest singular value.

B. Quantum One-Time Pad (QOTP)

Quantum one-time pad encryption (QOTP) [27]. The quantum one-time pad encryption (QOTP) perfectly hides any quantum message encrypted using a classical key chosen with uniform distribution.

The QOTP = (QOTPEnc, QOTPDdec) is defined as follows:

$$\begin{aligned} \text{QOTPEnc} : (\mathbb{C}^2)^{\otimes n} \times \{0, 1\}^{2n} &\longrightarrow (\mathbb{C}^2)^{\otimes n} \\ \text{and QOTPDdec} : (\mathbb{C}^2)^{\otimes n} \times \{0, 1\}^{2n} &\longrightarrow (\mathbb{C}^2)^{\otimes n} \end{aligned}$$

with

$$\begin{aligned} \text{QOTPEnc}(\rho, k) &= (X^{k_1} Z^{k_2} \otimes \dots \otimes X^{k_{2i-1}} Z^{k_{2i}} \otimes \dots \otimes X^{k_{2n-1}} Z^{k_{2n}})(\rho), \\ \text{QOTPDdec}(\rho, k) &= (Z^{k_2} X^{k_1} \otimes \dots \otimes Z^{k_{2i}} X^{k_{2i-1}} \otimes \dots \otimes Z^{k_{2n}} X^{k_{2n-1}})(\rho), \end{aligned}$$

for any message $\rho \in (\mathbb{C}^2)^{\otimes n}$ and key $k \in \{0, 1\}^{2n}$, where X^i, Z^i represents the quantum operation which is applied on the standard Pauli gates X, Z respectively to the i -th qubit.

The QOTP guarantees perfect secrecy

$$\mathbb{E}_{a,b}[\text{QOTP}_{a,b}(\varphi)] = \frac{1}{2^{2n}} \mathbb{I}. \quad (1)$$

This security extends to scenarios where the message is entangled with an external environment E . For any joint state $\varphi_{ME} \in \mathcal{D}(\mathcal{H}_{2^n} \otimes \mathcal{H}_E)$, we have

$$\mathbb{E}_{a,b}[(\text{QOTP}_{a,b} \otimes \mathbb{I}_E)(\varphi_{ME})] = \frac{1}{2^{2n}} \mathbb{I} \otimes \varphi_E.$$

Lemma 1. ([27], [28]) *The quantum one-time pad encryption scheme is correct and perfectly secure for a randomly chosen key. That is,*

$$\text{QOTPDdec}(\text{QOTPEnc}(\rho, k), k) = \rho$$

for any key $k \in \{0, 1\}^{2n}$, and

$$\sum_{k \in \{0, 1\}^{2n}} \frac{1}{2^{2n}} \text{QOTPEnc}(\rho, k) = \sum_{k \in \{0, 1\}^{2n}} \frac{1}{2^{2n}} \text{QOTPEnc}(\sigma, k)$$

for any two quantum states $\rho, \sigma \in (\mathbb{C}^2)^{\otimes n}$.

C. Broadcast encryption

We refer to [3] for a very good survey on broadcast encryption.

Definition 1. [3] A **broadcast-encryption scheme** is a tuple of three algorithms:

(Setup, Broadcast, Decrypt)

such that

- **Setup** : The algorithm accepts a user $u \in T$ and generates private information $P_u \in \mathcal{P}$ for that user.
- **Broadcast** : The algorithm takes a set of revoked users $\mathcal{R} \subseteq T$ and a session key $k \in \mathcal{K}$, and returns a broadcast message $B \in \mathcal{B}$ intended for distribution to all users.
- **Decrypt** : The algorithm is used by any user $u \in T$. It takes inputs (B, P_u, u) and attempts to recover the session key k . Decryption is successful if $u \notin \mathcal{R}$; otherwise, it fails.

D. Attribute-Based Encryption

The basic CP-ABE model includes four components: cloud service provider (CSP), attribute authority (AA), data owner (DO), and data user (DU).

Let $\mathcal{H}_M$ be the Hilbert space representing the message space, and let $\mathcal{D}(\mathcal{H})$ denote the set of density operators on a Hilbert space $\mathcal{H}$ defined by

$$\mathcal{D}(\mathcal{H}) := \{\rho \in \mathcal{L}(\mathcal{H}) \mid \rho \geq 0, \text{Tr}(\rho) = 1\}.$$

Let Ω be a finite set of attributes. An access policy is a monotone predicate $\text{Policy}: 2^\Omega \rightarrow \{0, 1\}$. For a set S , we write $S \models \text{Policy}$ if the policy is satisfied.

Definition 2 (Attribute-Based Quantum Encryption (ABQE)). A Attribute-Based Quantum Encryption scheme is a quadruple of quantum polynomial-time (QPT) algorithms:

ABQE = (Setup, KeyGen, Enc, Dec),

defined as follows:

- **Setup**: This is a QPT algorithm that takes a security parameter $\lambda \in \mathbb{N}$, and outputs public parameters PP and a master secret key MSK.

$$(\text{PP}, \text{MSK}) \leftarrow \text{Setup}(1^\lambda),$$

where $\lambda \in \mathbb{N}$ is a security parameter, PP denotes public parameters, and MSK is the master secret key.

- **Key Generation**: The key generation algorithm is a QPT algorithm defined as follows:

$$k_S \leftarrow \text{KeyGen}(\text{MSK}, S),$$

where $S \subseteq \Omega$ is an attribute set, and k_S is a secret key associated to S .

•Encryption:

$$C \leftarrow \text{Enc}(\text{PP}, \rho_M, \text{Policy}),$$

where $\rho_M \in \mathcal{D}(\mathcal{H}_M)$ is a quantum message, $\text{Policy}: 2^\Omega \rightarrow \{0, 1\}$, and the ciphertext $C \in \mathcal{D}(\mathcal{H}_C)$ encodes both the encrypted message and policy.

•Decryption:

$$\rho'_M \leftarrow \text{Dec}(C, k_S),$$

where $\rho'_M \in \mathcal{D}(\mathcal{H}_M)$ is the decryption attempt. If $S \models \text{Policy}$, then $\Delta(\rho'_M, \rho_M) < \varepsilon(\lambda)$; otherwise ρ'_M is independent of ρ_M .

•**Correctness**. Let $\varepsilon(\lambda)$ be a negligible function. Then for all $\lambda \in \mathbb{N}$, all $\rho_M \in \mathcal{D}(\mathcal{H}_M)$, all policies Policy, and all sets $S \subseteq \Omega$ such that $S \models \text{Policy}$, it holds that:

$$\Pr[\Delta(\text{Dec}(\text{Enc}(\text{PP}, \rho_M, \text{Policy}), k_S), \rho_M) \leq \varepsilon(\lambda)] \geq 1 - \varepsilon(\lambda),$$

where $\Delta(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1$ is the trace distance.

•**Security**. (Quantum Indistinguishability under Chosen Plaintext Attacks (QIND-CPA)). A QABE scheme is QIND-CPA secure if for all QPT adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$, the following experiment yields advantage at most negligible in λ .

• **QIND-CPA Security Experiment**: Let $b \in \{0, 1\}$ be chosen uniformly at random.

- 1) Challenger runs $(\text{PP}, \text{MSK}) \leftarrow \text{Setup}(1^\lambda)$, and gives PP to $\mathcal{A}_1$.
- 2) $\mathcal{A}_1$ outputs: $(\rho_0, \rho_1, \text{Policy}, \Omega)$, where $\rho_0, \rho_1 \in \mathcal{D}(\mathcal{H}_M)$ are of equal dimension, Policy is an access policy, and $S \subseteq 2^\Omega$ is a collection of attribute sets such that for all $S \subseteq \Omega$, we have $S \not\models \text{Policy}$.
- 3) For each $S \subseteq \Omega$, the challenger computes $k_S \leftarrow \text{KeyGen}(\text{MSK}, S)$ and gives all $\{k_S\}_{S \subseteq \Omega}$ to $\mathcal{A}_2$.
- 4) The challenger computes $C^* \leftarrow \text{Enc}(\text{PP}, \rho_b, \text{Policy})$ and gives C^* to $\mathcal{A}_2$.
- 5) $\mathcal{A}_2$ outputs a guess $b' \in \{0, 1\}$.

Define the adversary's advantage as

$$\text{Adv}_{\mathcal{A}}^{\text{QIND-CPA}}(\lambda) := \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

The scheme is secure if

$$\forall \text{ QPT } \mathcal{A}, \quad \text{Adv}_{\mathcal{A}}^{\text{QIND-CPA}}(\lambda) < \varepsilon(\lambda).$$

III. POLICY-BASED QUANTUM BROADCAST ENCRYPTION

In this section, we first define a policy tree, where each leaf node is defined by a unique policy and each policy is defined by an attribute-based access tree. Then we have constructed a general policy-based quantum broadcast encryption with multiple messages encrypted simultaneously by traversing the policy-based access tree. For our construction of policy-based broadcast encryption, we first review some definitions of access trees with threshold gates and access trees with AND-OR gates.

Definition 3 (Threshold-Gate Access Tree). ([29]) A Threshold-Gate Access Tree is a rooted tree $T = (V, E)$ where

each internal node $v \in V$ is associated with a set of children $\text{child}(v) = \{v_1, v_2, \dots, v_{n_v}\}$, where $n_v \geq 2$, and a threshold value $t_v \in \{1, 2, \dots, n_v\}$. Each leaf node $v \in L \subset V$ is labeled by an attribute $\alpha \in \Omega$, where Ω is the set of all attributes.

An attribute set $S \subseteq \Omega$ is said to satisfy the access tree T if, by inductively declaring a leaf node with label α is satisfied by S if $\alpha \in S$, an internal node v is satisfied by S if at least t_v of its children are satisfied, the root of T is satisfied.

Definition 4. [29] Let Ω be a finite set of attributes. An **AND-OR access tree** is a finite rooted tree $T = (N, r, \text{child}, \ell)$, where

- N is the finite set of nodes;
- $r \in N$ is the distinguished root node;
- $\text{child} : N \rightarrow 2^N$ is the child function, satisfying:

$$\forall v \in N, \text{ if } \text{child}(v) = \emptyset, \text{ then } v \text{ is a leaf};$$

- $\ell : N \rightarrow \Omega \cup \{\text{AND}, \text{OR}\}$ satisfies:

$$\ell(v) = \begin{cases} \alpha \in \Omega, & \text{if } v \text{ is a leaf,} \\ \text{AND or OR,} & \text{if } v \text{ is internal.} \end{cases}$$

Given $S \subseteq \Omega$, define satisfaction recursively

$$\text{Sat}_S(v) := \begin{cases} 1, & \text{if } v \text{ is a leaf and } \ell(v) \in S, \\ 1, & \text{if } \ell(v) = \text{AND and } \forall w \in \text{child}(v) : \\ & \text{Sat}_S(w) = 1, \\ 1, & \text{if } \ell(v) = \text{OR and } \exists w \in \text{child}(v) : \\ & \text{Sat}_S(w) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

S satisfies T iff $\text{Sat}_S(r) = 1$.

For our construction of policy-based broadcast encryption, we define the policy tree as follows.

When we have N_L distinct messages, and each message $\rho_{(0)}^{(k)}$ is governed by an independent policy Policy_k such that each Policy_k is defined by its own AND-OR attribute-based access tree $T^{(k)}$ with a logical formula over attributes, we construct a single policy tree T_{Policy} that provides N_L unique root-to-leaf paths, where the k -th path P_k is associated with Policy_k .

Let $\{\text{Policy}_1, \dots, \text{Policy}_{N_L}\}$ be the set of N_L access policies, where each Policy_k defines the conditions for accessing message $\rho_{(0)}^{(k)}$.

Now we define an analogous AND-OR policy access tree, which we will use in our construction.

Definition 5 (AND-OR Policy Access Tree). Let $\{\text{Policy}_k\}_{k=1}^{N_L}$ be a set of attribute-based access policies, where each Policy_k is defined by its own access tree $T^{(k)}$. An AND-OR Policy Access Tree is a rooted tree $T_{\text{Policy}} = (N', r', \text{child}, \ell)$, where:

- N' is a finite set of nodes;
- $r' \in N'$ is the root node;
- $\text{child} : N' \rightarrow 2^{N'}$ is a child function;

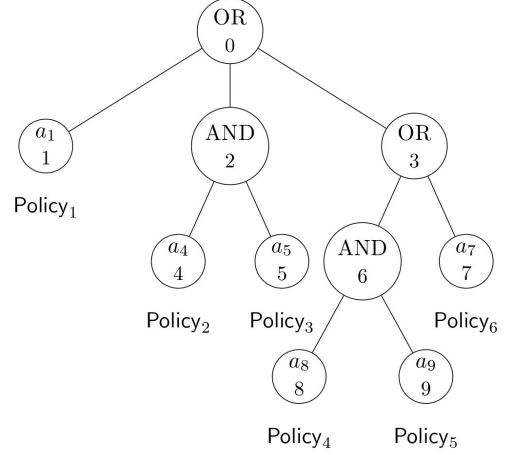


Fig. 1. An AND-OR policy access tree T_{Policy} with 6 policies.

- $\ell : N' \rightarrow \{\text{AND}, \text{OR}\} \cup \{\text{Policy}_1, \dots, \text{Policy}_{N_L}\}$ is a labeling function such that:

$$\ell(v) = \begin{cases} \text{Policy}_k, & \text{if } \text{child}(v) = \emptyset, \\ \text{AND or OR,} & \text{otherwise.} \end{cases}$$

Each policy Policy_k is evaluated as described above, using its own tree $T^{(k)}$, and attribute set $S \subseteq \Omega$.

For a leaf node labeled with Policy_k , we define:

$$\text{Sat}_S(v) := 1 \quad \text{iff} \quad S \models T^{(k)}.$$

Then recursively we define

$$\text{Sat}_S(v) := \begin{cases} 1, & \text{if } \ell(v) = \text{Policy}_k \text{ and } S \models T^{(k)}, \\ 1, & \ell(v) = \text{AND} \wedge \forall w \in \text{child}(v), \text{Sat}_S(w) = 1, \\ 1, & \ell(v) = \text{OR} \wedge \exists w \in \text{child}(v), \text{Sat}_S(w) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

We say S satisfies the policy tree T_{Policy} iff $\text{Sat}_S(r) = 1$.

Within the context of attribute-based access control (ABAC) or policy-based systems, a Policy can be represented by an AND-OR tree. Here in our construction, each policy Policy_k represents an AND-OR access tree.

Here is an example of an AND-OR policy tree in T_{Policy} with policies $\text{Policy}_1, \dots, \text{Policy}_6$.

A. Main construction

Let $T = (V, E)$ be a finite, rooted tree, where V is the set of vertices and E is the set of edges. Let $r \in V$ be the root node. Let $L \subset V$ be the set of leaf nodes. The number of leaf nodes is $N_L = |L|$. For $k = 1, \dots, N_L$, there exists a unique simple path P_k from the root r to each leaf node $l_k \in L$. This path is represented as an ordered sequence of vertices

$$P_k : v_0^{(k)} \rightarrow v_1^{(k)} \rightarrow \dots \rightarrow v_{d_k}^{(k)},$$

where $v_0^{(k)} = r$ is the root, $v_{d_k}^{(k)} = l_k$ is the k -th leaf node, and $(v_j^{(k)}, v_{j+1}^{(k)}) \in E$ for all $j \in \{0, \dots, d_k - 1\}$. The length of path

P_k , i.e., the number of edges constituting the path P_k , is d_k . The maximum depth of the tree is $D = \max_{k \in \{1, \dots, N_L\}} \{d_k\}$. For notational convenience, if $j > d_k$, we define $v_j^{(k)} = v_{d_k}^{(k)} = l_k$. This means a path, once it reaches its leaf, is considered to stay at that leaf for subsequent levels up to D . This is because even if we reached a leaf node of a path, we still traverse the other longer paths where the walk stays on that leaf node.

The total quantum system comprises N_L quantum subsystems, each associated with a unique root-to-leaf path P_k .

Path State Space. For each path P_k , we define a Hilbert space $\mathcal{H}_{P_k} \cong \mathbb{C}^{|V|}$ spanned by the orthonormal basis $\{|v\rangle_{P_k} : v \in V\}$. The state $|v\rangle_{P_k}$ signifies that the k -th path component is currently associated with vertex v . We often write $|v\rangle$ when the context P_k is clear. A projector onto the state associated with vertex v in $\mathcal{H}_{P_k}$ is $\Pi_v^{(P_k)} = |v\rangle\langle v|_{P_k}$.

Message State Space. For each path P_k , we define a Hilbert space $\mathcal{H}_{M_k}$ for the k -th quantum message $\rho^{(k)}$. If messages are composed of n qubits, then $\mathcal{H}_{M_k} \cong (\mathbb{C}^2)^{\otimes n}$, and $\dim(\mathcal{H}_{M_k}) = 2^n$.

Total Hilbert Space. The Hilbert space for the entire system is the tensor product of the spaces for all paths

$$\mathcal{H}_B := \bigotimes_{k=1}^{N_L} (\mathcal{H}_{P_k} \otimes \mathcal{H}_{M_k}).$$

The dimension of $\mathcal{H}_B$ is $(|V| \cdot 2^n)^{N_L}$ if all message spaces have dimension 2^n .

Initial State of the System. Let $\mathcal{D}(\mathcal{H}_{M_k})$ be the set of density operators defined on the message space $\mathcal{H}_{M_k}$. The system is initialized with N_L distinct quantum messages $\rho_{(0)}^{(k)} \in \mathcal{D}(\mathcal{H}_{M_k})$, each to be associated with path P_k . Initially, all path registers are set to the root node r . The total initial state $\rho_B^{(0)} \in \mathcal{D}(\mathcal{H}_B)$ is

$$\rho_B^{(0)} = \bigotimes_{k=1}^{N_L} \left(\Pi_r^{(P_k)} \otimes \rho_{(0)}^{(k)} \right),$$

where $\Pi_r^{(P_k)} = |r\rangle\langle r|_{P_k}$ is the projector for path k 's register onto the root state, and $\rho_{(0)}^{(k)}$ is the initial density operator for the k -th message, acting on $\mathcal{H}_{M_k}$.

Encryption Process. The encryption process evolves the total state $\rho_B^{(j)}$ through D levels, from $j = 0$ to $j = D - 1$. Each level j corresponds to processing the tree edges at depth j .

Now we define the operators for encryption at level j . For each path $P_k : v_0^{(k)} \rightarrow v_1^{(k)} \rightarrow \dots \rightarrow v_{d_k}^{(k)}$ and current processing level $j \in \{0, \dots, D - 1\}$,

- current node for path P_k : $u_k := v_j^{(k)}$.
- target node for path P_k for this step: $w_k := v_{j+1}^{(k)}$, as $v_{j+1}^{(k)} = v_{d_k}^{(k)} = l_k$ if $j \geq d_k$.

Path Transition Operator. For path P_k , the transition from node u_k to w_k is implemented by a unitary operator $P_{u_k \leftrightarrow w_k}^{(P_k)}$ acting on $\mathcal{H}_{P_k}$. We choose the following operator

$$P_{u_k \leftrightarrow w_k}^{(P_k)} := \begin{cases} |w_k\rangle_{P_k} \langle u_k|_{P_k} + |u_k\rangle_{P_k} \langle w_k|_{P_k} \\ + \sum_{v \in V \setminus \{u_k, w_k\}} |v\rangle_{P_k} \langle v|_{P_k} & \text{if } u_k \neq w_k \\ \mathbb{I}_{P_k} & \text{if } u_k = w_k \end{cases}$$

Policy-Based Quantum Broadcast Encryption

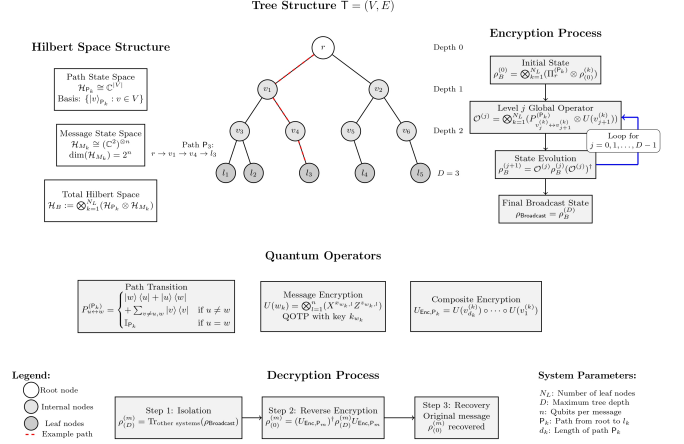


Fig. 2. Quantum Broadcast Encryption based on traversing tree and composite state construction

If $u_k = w_k$, which occurs if $j \geq d_k$, then $P_{u_k \leftrightarrow w_k}^{(P_k)} = \mathbb{I}_{P_k}$ the identity operator on $\mathcal{H}_{P_k}$. This operator ensures $P_{u_k \leftrightarrow w_k}^{(P_k)} \Pi_{u_k}^{(P_k)} \left(P_{u_k \leftrightarrow w_k}^{(P_k)} \right)^\dagger = \Pi_{w_k}^{(P_k)}$.

Circuit construction for path transition operator. Let $\mathcal{H}_V = (\mathbb{C}^2)^{\otimes m}$, where $m = \lceil \log_2 |V| \rceil$. Each vertex $v \in V$ is encoded as a computational basis state $|v\rangle \equiv |v_{m-1} \dots v_0\rangle$. We use an auxiliary qubit in the space $\mathcal{H}_{\text{aux}} = \mathbb{C}^2$, initialized to $|0\rangle$.

For a vertex v with binary representation $(v_{m-1}, \dots, v_0)$, the detector D_v is defined as:

$$D_v = \left(\bigotimes_{i=0}^{m-1} X^{1-v_i} \right) \cdot C^m(X)_{[0, \dots, m-1] \rightarrow \text{aux}} \cdot \left(\bigotimes_{i=0}^{m-1} X^{1-v_i} \right) \quad (2)$$

where $C^m(X)$ is the m -qubit controlled-NOT gate targeting the auxiliary qubit.

The detector operator D_v that controllably flips the auxiliary qubit if and only if the vertex register is in the state $|v\rangle$. The operator D_v acts on a state $|\psi\rangle |a\rangle$ where $|\psi\rangle \in \mathcal{H}_V$ and $|a\rangle \in \mathcal{H}_{\text{aux}}$ as follows:

$$D_v(|\psi\rangle \otimes |a\rangle) = |\psi\rangle \otimes |a + \delta_{\psi, v}\rangle \quad (3)$$

where $\delta_{\psi, v}$ is the Kronecker delta. The X gates transform the state $|v\rangle$ to $|1\rangle \equiv |11 \dots 1\rangle$ to activate the $C^m(X)$ gate, and then transform the state back, leaving the vertex register $|\psi\rangle$ unchanged.

Let $\mathcal{V} = \{i \mid u_i \neq w_i\}$ be the set of indices where the binary representations of u and w differ. Consider the following controlled-swap operator $\mathcal{T}_{\mathcal{V}} = \prod_{i \in \mathcal{V}} \text{CNOT}_{\text{aux} \rightarrow i}$. When the auxiliary qubit is $|1\rangle$, this operator flips the target qubits in the vertex register at all positions $i \in \mathcal{V}$. This action transforms $|u\rangle$ to $|w\rangle$ and $|w\rangle$ to $|u\rangle$ simultaneously, that is, $\mathcal{T}_{\mathcal{V}}(|u\rangle \otimes |1\rangle) = |w\rangle \otimes |1\rangle$, $\mathcal{T}_{\mathcal{V}}(|w\rangle \otimes |1\rangle) = |u\rangle \otimes |1\rangle$. If the auxiliary qubit is $|0\rangle$, no action is taken, that is $\mathcal{T}_{\mathcal{V}}(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |0\rangle$. The path transition operator $P_{u \leftrightarrow w}$ is implemented by the unitary operator $\tilde{P}_{u \leftrightarrow w}$ acting on $\mathcal{H}_V \otimes \mathcal{H}_{\text{aux}}$, defined by, $\tilde{P}_{u \leftrightarrow w} := (D_u D_w) \cdot \mathcal{T}_{\mathcal{V}} \cdot (D_u D_w)$ such that for any vertex state $|v\rangle$, the final state is $(P_{u \leftrightarrow w} |v\rangle) \otimes |0\rangle$.

Message Encryption Unitary. For path P_k , when transitioning from u_k to w_k , a unitary operator $U(w_k)$ acts on the message space $\mathcal{H}_{M_k}$. This unitary is associated with the target node w_k of the current step (e.g., derived from a key k_{w_k} specific to the node w_k). If $u_k = w_k$ (i.e., $j \geq d_k$, path P_k has already reached its leaf and remains there), then $U(w_k) = \mathbb{I}_{M_k}$, i.e., no further encryption on this message. Here in our construction for $U(w_k)$ using a key $k_{w_k} = (x_{w_k}, z_{w_k})$, we define $U(w_k)$ as a QOTP operator $U(w_k) = \bigotimes_{l=1}^n (X^{x_{w_k,l}} Z^{z_{w_k,l}})$, where X, Z are Pauli operators on the l -qubit of the message in $\mathcal{H}_{M_k}$.

Now we present the general quantum broadcast encryption algorithm based on the policy access tree traversal and encrypting multiple messages simultaneously [See Algorithm 1, Page 10].

Description of the construction of policy-based quantum broadcast encryption

State Transformation per Level

Global Encryption Operator for Level j :

The global encryption operator $\mathcal{O}^{(j)}$ that advances the system state from level j to $j+1$ is the tensor product of individual operators for each path

$$\mathcal{O}^{(j)} := \bigotimes_{k=1}^{N_L} \left(P_{v_j^{(k)} \leftrightarrow v_{j+1}^{(k)}}^{(P_k)} \otimes U(v_{j+1}^{(k)}) \right).$$

Note that $v_j^{(k)}$ is the node of path k at depth j , and $v_{j+1}^{(k)}$ is the node of path k at depth $j+1$. If path k is shorter than $j+1$ (i.e., $d_k \leq j$), then $v_j^{(k)} = v_{j+1}^{(k)} = l_k$, $P^{(P_k)} = \mathbb{I}_{P_k}$ and $U(v_{j+1}^{(k)}) = \mathbb{I}_{M_k}$.

The total state of the system is updated iteratively for $j = 0, \dots, D-1$,

$$\rho_B^{(j+1)} = \mathcal{O}^{(j)} \rho_B^{(j)} (\mathcal{O}^{(j)})^\dagger.$$

Expanding this, $\rho_B^{(j+1)}$ we get,

$$\begin{aligned} \rho_B^{(j+1)} &= \bigotimes_{k=1}^{N_L} \left[\left(P_{v_j^{(k)} \leftrightarrow v_{j+1}^{(k)}}^{(P_k)} \Pi_{v_j^{(k)}}^{(P_k)} \left(P_{v_j^{(k)} \leftrightarrow v_{j+1}^{(k)}}^{(P_k)} \right)^\dagger \right) \right. \\ &\quad \left. \otimes \left(U(v_{j+1}^{(k)}) \rho_{(j)}^{(k)} U(v_{j+1}^{(k)})^\dagger \right) \right] \\ &= \bigotimes_{k=1}^{N_L} \left(\Pi_{v_{j+1}^{(k)}}^{(P_k)} \otimes \rho_{(j+1)}^{(k)} \right) \end{aligned}$$

where $\rho_{(j)}^{(k)}$ is the message state of the k -th path after j levels of encryption, and $\rho_{(j+1)}^{(k)} = U(v_{j+1}^{(k)}) \rho_{(j)}^{(k)} U(v_{j+1}^{(k)})^\dagger$.

Final Encrypted State: After D levels of encryption (i.e., iterating j from 0 to $D-1$), the final ciphertext is

$$\rho_{\text{Broadcast}} \equiv \rho_B^{(D)}.$$

This final state for broadcasting is defined by

$$\rho_{\text{Broadcast}} = \bigotimes_{k=1}^{N_L} \left(\Pi_{l_k}^{(P_k)} \otimes \rho_{(D)}^{(k)} \right),$$

where $l_k = v_{d_k}^{(k)}$ is the leaf node of path P_k . The fully encrypted k -th message component $\rho_{(D)}^{(k)}$ is

$$\rho_{(D)}^{(k)} = U_{\text{Enc}, P_k} \rho_{(0)}^{(k)} (U_{\text{Enc}, P_k})^\dagger,$$

where U_{Enc, P_k} is the composite unitary operator resulting from the sequence of unitaries applied along path P_k :

$$U_{\text{Enc}, P_k} = U(v_{d_k}^{(k)}) \circ U(v_{d_k-1}^{(k)}) \circ \dots \circ U(v_1^{(k)}).$$

Note that $U(v_j^{(k)})$ is the unitary associated with the j -th node visited on path P_k , excluding the root $v_0^{(k)}$. The unitaries are applied in sequence $U(v_1^{(k)})$ first, then $U(v_2^{(k)})$, up to $U(v_{d_k}^{(k)})$.

Decryption Process: To decrypt a specific message, say $\rho_{(0)}^{(m)}$ associated with path P_m , from the final broadcast ciphertext $\rho_{\text{Broadcast}}$. The path is

$$P_m : v_0^{(m)} \rightarrow v_1^{(m)} \rightarrow \dots \rightarrow v_{d_m}^{(m)} = l_m.$$

Step 1: Isolation of the Target Encrypted Message Component:

The encrypted message component $\rho_{(D)}^{(m)}$ must first be isolated from $\rho_{\text{Broadcast}}$. This is achieved by performing a partial trace over all other path systems (path and message registers for $k \neq m$) and over the path register of the m -th system $\mathcal{H}_{P_m}$.

Isolation via Partial Trace. The isolated encrypted message component $\rho_{(D)}^{(m)} \in \mathcal{D}(\mathcal{H}_{M_m})$ is obtained as

$$\rho_{(D)}^{(m)} = \text{Tr}_{(\bigotimes_{k \neq m} (\mathcal{H}_{P_k} \otimes \mathcal{H}_{M_k})) \otimes \mathcal{H}_{P_m}} (\rho_{\text{Broadcast}}).$$

Given the tensor product structure of $\rho_{\text{Broadcast}}$

$$\rho_{\text{Broadcast}} = \left(\Pi_{l_m}^{(P_m)} \otimes \rho_{(D)}^{(m)} \right) \otimes \left(\bigotimes_{k \neq m} \left(\Pi_{l_k}^{(P_k)} \otimes \rho_{(D)}^{(k)} \right) \right),$$

the partial trace becomes

$$\begin{aligned} \rho_{(D)}^{(m)} &= \left(\text{Tr}_{\mathcal{H}_{P_m}} [\Pi_{l_m}^{(P_m)}] \right) \rho_{(D)}^{(m)} \\ &\quad \cdot \prod_{k \neq m} \left(\text{Tr}_{\mathcal{H}_{P_k} \otimes \mathcal{H}_{M_k}} [\Pi_{l_k}^{(P_k)} \otimes \rho_{(D)}^{(k)}] \right) \\ &= (1) \cdot \rho_{(D)}^{(m)} \cdot \prod_{k \neq m} \left(\text{Tr}_{\mathcal{H}_{P_k}} [\Pi_{l_k}^{(P_k)}] \cdot \text{Tr}_{\mathcal{H}_{M_k}} [\rho_{(D)}^{(k)}] \right) \\ &= \rho_{(D)}^{(m)} \cdot \prod_{k \neq m} (1 \cdot 1) = \rho_{(D)}^{(m)}. \end{aligned}$$

This confirms that the operation correctly extracts the desired encrypted message component.

Step 2: Reversing Unitary Transformations: To recover the original message $\rho_{(0)}^{(m)}$ from the isolated encrypted component $\rho_{(D)}^{(m)}$, we must apply the adjoints of the unitary operators used during encryption for path P_m , in the reverse order of their application.

The composite encryption unitary for path P_m was

$$U_{\text{Enc}, P_m} = U(v_{d_m}^{(m)}) \circ U(v_{d_m-1}^{(m)}) \circ \dots \circ U(v_1^{(m)}).$$

The original message is recovered by the transformation

$$\rho_{(0)}^{(m)} = (U_{\text{Enc}, P_m})^\dagger \rho_{(D)}^{(m)} U_{\text{Enc}, P_m}.$$

The adjoint $(U_{\text{Enc}, P_m})^\dagger$ is

$$(U_{\text{Enc}, P_m})^\dagger = (U(v_1^{(m)}))^\dagger \circ (U(v_2^{(m)}))^\dagger \circ \dots \circ (U(v_{d_m}^{(m)}))^\dagger.$$

Therefore, by decryption, we get

$$\begin{aligned} \rho_{(0)}^{(m)} &= \left((U(v_1^{(m)}))^\dagger \dots (U(v_{d_m}^{(m)}))^\dagger \right) \\ &\quad \rho_{(D)}^{(m)} \left(U(v_{d_m}^{(m)}) \dots U(v_1^{(m)}) \right). \end{aligned}$$

This decryption can be performed iteratively. Let $\tilde{\rho}_{M_m}^{(d_m)} = \rho_{(D)}^{(m)}$. For $j = d_m, d_m - 1, \dots, 1$ (iterating downwards over the nodes $v_j^{(m)}$ whose unitaries were applied):

$$\tilde{\rho}_{M_m}^{(j-1)} = (U(v_j^{(m)}))^\dagger \tilde{\rho}_{M_m}^{(j)} U(v_j^{(m)}).$$

The final result of this iterative process is $\rho_{(0)}^{(m)} = \tilde{\rho}_{M_m}^{(0)}$.

Remark 1. • Path transition operators, along with routing and traversal mechanisms, play a crucial role in tracking the sequence of vertices along specific paths and in determining when leaf nodes are reached.

• The path transition operators $P_{u \leftrightarrow w}^{(P_k)}$ do not need to be explicitly reversed during the decryption of the message component $\rho_{(D)}^{(m)}$. This is because these operators act on the path state spaces $\mathcal{H}_{P_k}$, which are traced out during the isolation step. The decryption process focuses solely on reversing transformations on the message state in $\mathcal{H}_{M_m}$.

For a rigorous example of our construction, we refer to the supplementary document.

Note: To share the secret keys SK_{user} that depend on each vertex on a particular path associated with the particular policies, we may use the classical secret-sharing scheme based on Shamir's secret-sharing scheme [30].

Correctness. For any authorized user with access to the path P_m , the decryption algorithm correctly recovers the original message $\rho_{(0)}^{(m)}$ from $\rho_{\text{Broadcast}}$.

Consider an authorized user for path $P_m : v_0^{(m)} \rightarrow \dots \rightarrow v_{d_m}^{(m)} = l_m$ with secret key SK_{user} enabling decryption of leaf l_m . The broadcast state is

$$\rho_{\text{Broadcast}} = \bigotimes_{k=1}^{N_L} \left(\Pi_{l_k}^{(P_k)} \otimes \rho_{(D)}^{(k)} \right),$$

where $\Pi_{l_k}^{(P_k)} = |l_k\rangle \langle l_k|_{P_k}$.

First, the target component is separated by partial trace

$$\rho_m = \text{Tr}_{\bigotimes_{k \neq m} (\mathcal{H}_{P_k} \otimes \mathcal{H}_{M_k})} (\rho_{\text{Broadcast}}) = \Pi_{l_m}^{(P_m)} \otimes \rho_{(D)}^{(m)}$$

Then the user applies unitary operators for decryption. Since $\rho_{(D)}^{(m)} = U_{\text{Enc}, P_m} \rho_{(0)}^{(m)} (U_{\text{Enc}, P_m})^\dagger$ where $U_{\text{Enc}, P_m} = U(v_{d_m}^{(m)}) \circ \dots \circ U(v_1^{(m)})$, the authorized user reconstructs keys $\{k_{v_j^{(m)}}\}$ via Shamir's scheme [30] and computes

$$\tilde{\rho}_{(0)}^{(m)} = (U_{\text{Enc}, P_m})^\dagger \rho_{(D)}^{(m)} U_{\text{Enc}, P_m} = \rho_{(0)}^{(m)}$$

since $(U_{\text{Enc}, P_m})^\dagger U_{\text{Enc}, P_m} = \mathbb{I}_{M_m}$.

Throughout the encryption process, path registers remain in product states. This can be proved by induction on encryption level j .

Base case ($j = 0$): $\rho_B^{(0)} = \bigotimes_{k=1}^{N_L} \left(|r\rangle \langle r|_{P_k} \otimes \rho_{(0)}^{(k)} \right)$ is clearly a product state.

Inductive step: Assume

$$\rho_B^{(j)} = \bigotimes_{k=1}^{N_L} \left(|v_j^{(k)}\rangle \langle v_j^{(k)}|_{P_k} \otimes \rho_{(j)}^{(k)} \right).$$

Applying $\mathcal{O}^{(j)} = \bigotimes_{k=1}^{N_L} \left(P_{v_j^{(k)} \leftrightarrow v_{j+1}^{(k)}}^{(P_k)} \otimes U(v_{j+1}^{(k)}) \right)$, we have,

$$\begin{aligned} \rho_B^{(j+1)} &= \mathcal{O}^{(j)} \rho_B^{(j)} (\mathcal{O}^{(j)})^\dagger \\ &= \bigotimes_{k=1}^{N_L} \left(|v_{j+1}^{(k)}\rangle \langle v_{j+1}^{(k)}|_{P_k} \otimes \rho_{(j+1)}^{(k)} \right) \end{aligned}$$

The tensor product structure is preserved by the construction of $\mathcal{O}^{(j)}$.

• **Correlation Analysis.** Path registers of different paths never become entangled during encryption.

The encryption operator has the structure

$$\mathcal{O}^{(j)} = \bigotimes_{k=1}^{N_L} \mathcal{O}_k^{(j)} \quad (4)$$

where $\mathcal{O}_k^{(j)} = P_{v_j^{(k)} \leftrightarrow v_{j+1}^{(k)}}^{(P_k)} \otimes U(v_{j+1}^{(k)})$ acts only on the k -th path-message subsystem. For any $j \in [D]$, define

$$\rho_B^{(j)} = \bigotimes_{k=1}^{N_L} \left(|v_{\min(j, d_k)}^{(k)}\rangle \langle v_{\min(j, d_k)}^{(k)}|_{P_k} \otimes \rho_{(j)}^{(k)} \right), \quad (5)$$

where $\rho_{(j)}^{(k)} = U_{v_{\min(j, d_k)}^{(k)}} \dots U_{v_1^{(k)}} \rho_{(0)}^{(k)} (U_{v_1^{(k)}})^\dagger \dots (U_{v_{\min(j, d_k)}^{(k)}})^\dagger$. Then, for any $j \in [D]$ and distinct $k, k' \in [N_L]$,

$$\text{Tr}_{\{k, k'\}} [\rho_B^{(j)}] = \rho_{(j)}^{(k)} \otimes \rho_{(j)}^{(k')}. \quad (6)$$

This tensor product decomposition ensures that no correlations develop between subsystems corresponding to different paths.

Remark 2. There are many ways to reduce complexity by grouping paths passing through a particular vertex, but in that case, using a single key, other ciphertexts can also be decrypted. Therefore, we use different keys associated with each vertex for encryption based on QOTP.

Security Analysis: Information-Theoretic Security.

Perfect Secrecy. Under uniformly random QOTP keys, the broadcast encryption scheme provides perfect secrecy against any coalition of unauthorized recipients. For an adversary without knowledge of the path P_m 's keys,

$$I(\rho_{(0)}^{(m)} : \rho_{\text{Broadcast}}) = 0, \quad (7)$$

where $I(\cdot : \cdot)$ denotes quantum mutual information.

Consider any unauthorized coalition $\mathcal{S} \subset \{1, 2, \dots, N_L\}$ and any two collections of input messages $\{\rho_{(0)}^{(k)}\}_{k=1}^{N_L}$ and $\{\sigma_{(0)}^{(k)}\}_{k=1}^{N_L}$.

Let $\rho_{\text{Broadcast}}$ and $\sigma_{\text{Broadcast}}$ be the corresponding broadcast ciphertexts. The view of coalition $\mathcal{S}$ is obtained by tracing out all systems not accessible to $\mathcal{S}$

$$\rho_{\mathcal{S}} = \text{Tr}_{\mathcal{S}^c}[\rho_{\text{Broadcast}}] \quad (8)$$

$$\sigma_{\mathcal{S}} = \text{Tr}_{\mathcal{S}^c}[\sigma_{\text{Broadcast}}] \quad (9)$$

For any path $k \in \mathcal{S}^c$, the final encrypted message is

$$\rho_B^{(k)} = U_{P_k} \rho_{(0)}^{(k)} U_{P_k}^\dagger \quad (10)$$

where $U_{P_k} = U(v_{d_k}^{(k)}) \circ U(v_{d_k-1}^{(k)}) \circ \dots \circ U(v_1^{(k)})$.

Since each $U(v_i^{(k)})$ is a QOTP with uniformly random key, and the keys are independent across different nodes, by 1 and 1, we have

$$\mathbb{E}_{k_v}[\rho_B^{(k)}] = \mathbb{E}_{k_v}[\sigma_B^{(k)}] = \frac{\mathbb{I}}{2^n} \quad (11)$$

Therefore

$$\mathbb{E}_{k_v}[\rho_{\mathcal{S}}] = \mathbb{E}_{k_v}[\sigma_{\mathcal{S}}] \quad (12)$$

This proves perfect secrecy against any unauthorized coalition.

IV. SECURITY AGAINST ACTIVE QUANTUM ATTACKS

In this section, we discuss possible active attacks against both of our constructions.

• **Quantum Measurement Attacks.** Our scheme remains secure against adversaries who perform quantum measurements during the encryption process. Consider an adversary who intercepts and measures the state at level j ,

$$\rho_{(j)}^{(j)} = \bigotimes_{k=1}^{N_L} \left(\left| v_j^{(k)} \right\rangle \left\langle v_j^{(k)} \right|_{P_k} \otimes \rho_{(j)}^{(k)} \right) \quad (13)$$

If the adversary measures the path register in the computational basis, they learn the current position $v_j^{(k)}$ on each path. This reveals only partial path information on the position at level j . The full path P_k requires knowledge of all transitions, and without the complete path, the adversary cannot determine which leaf l_k will be reached.

If the adversary measures the message register at level j

$$\rho_{(j)}^{(k)} = U(v_j^{(k)}) \circ \dots \circ U(v_1^{(k)}) \rho_{(0)}^{(k)} U(v_1^{(k)})^\dagger \circ \dots \circ U(v_j^{(k)})^\dagger \quad (14)$$

Without knowing the keys $\{k_{v_1^{(k)}}, \dots, k_{v_j^{(k)}}\}$, the state appears maximally mixed

$$\mathbb{E}_{\text{keys}}[\rho_{(j)}^{(k)}] = \frac{\mathbb{I}_{M_k}}{2^n}. \quad (15)$$

Any measurement yields only random outcomes with probability $1/2^n$ for each basis state. If the adversary performs an entangling operation U_E with an ancilla system,

$$U_E : \rho_B^{(j)} \otimes |0\rangle \langle 0|_E \mapsto \rho_{BE} \quad (16)$$

the subsequent encryption operations act only on the system registers, not the adversary's ancilla. The final state maintains the tensor product structure, preventing information leakage through entanglement.

• **Trojan Horse Attacks.** The scheme is secure against Trojan horse attacks where an adversary injects malicious quantum

states. Consider an adversary who tries to inject a Trojan horse state $|\psi\rangle_T$ into the system. Suppose the adversary tries to inject the state $|\psi\rangle_T$ into the path register. The path transition operators are designed to work on computational basis states. Any superposition will be detected, as the detection circuits only activate for specific basis states. Superposition states will not trigger the correct detection pattern. The encryption will fail to proceed correctly, alerting to tampering.

If the adversary injects a known state $|\phi\rangle_T$ into the message register to trace encryption

$$\rho_{\text{trojan}}^{(k)} = |\phi\rangle_T \langle \phi|_T \quad (17)$$

After encryption with unknown keys

$$\rho_{\text{encrypted}}^{(k)} = U_{\text{Enc}, P_k} |\phi\rangle_T \langle \phi|_T U_{\text{Enc}, P_k}^\dagger. \quad (18)$$

Without knowing the keys, the adversary cannot distinguish this from any other encrypted state, as QOTP provides perfect randomization.

• **Man in the middle attack.** A Man-in-the-Middle (MITM) attack is an active attack where an adversary intercepts a communication channel, impersonating both legitimate parties to each other. The adversary can then read, modify, or inject messages into the channel without the legitimate parties' knowledge. An adversary who intercepts the broadcast state $\rho_{\text{Broadcast}}$ cannot learn the messages $\{\rho_{(0)}^{(k)}\}$ due to the perfect secrecy of QOTP, assuming the keys are unknown to them.

• **Quantum Side-Channel Attacks.** The scheme maintains security even under quantum side-channel attacks that may leak partial information about keys or operations. Consider side-channel leakage modeled as

$$\mathcal{E}_{\text{leak}}(\rho \otimes |k\rangle \langle k|) = \rho \otimes \mathcal{N}_k(|k\rangle \langle k|), \quad (19)$$

where $\mathcal{N}_k$ is a quantum channel that may leak information about key k . Each node uses an independent key, so leakage about $k_{v_j^{(k)}}$ provides no information about $k_{v_{j'}^{(k)'}}$ for $j' \neq j$. Compromise of keys at one level j doesn't compromise security at other levels, as each level uses fresh keys chosen with uniform randomness.

V. DISCUSSIONS

In this section, we discuss resource estimation for our encryption scheme.

Key Requirements (Total Key Length). The scheme requires

$$\sum_{v \in V \setminus \{r\}} \lceil \log_2 \dim(U_v) \rceil = (|V| - 1) \cdot 2n \text{ bits}, \quad (20)$$

for n -qubit messages encrypted using QOTP.

Qubit requirements. The total quantum state is held in the broadcast register $\mathcal{H}_B = \bigotimes_{k=1}^{N_L} (\mathcal{H}_{P_k} \otimes \mathcal{H}_{M_k})$.

Message Registers ($\mathcal{H}_{M_k}$). N_L messages of n qubits each require $N_L \cdot n$ qubits.

Path Registers ($\mathcal{H}_{P_k}$). To uniquely address $|V|$ nodes, each of the N_L path registers requires $m = \lceil \log_2 |V| \rceil$ qubits. This totals $N_L \cdot m$ qubits.

Auxiliary Qubits. The circuit for the path transition operator requires one auxiliary qubit per path. This adds N_L qubits.

The total number of qubits, $Q_{\text{Broadcast}}$, is

$$Q_{\text{Broadcast}} = N_L \cdot n + N_L \cdot \lceil \log_2 |V| \rceil + N_L = N_L(n + \lceil \log_2 |V| \rceil + 1) \quad [1]$$

For a reasonably balanced tree, where $|V| \in O(N_L)$, the qubit complexity is

$$Q_{\text{Broadcast}} = O(N_L(n + \log N_L))$$

Circuit depth. The encryption process has D sequential levels. The total depth is D times the depth of the operator $\mathcal{O}^{(j)}$ applied at each level. The message unitary $U(w_k)$ (QOTP) has a depth of 1. The path transition operator $\hat{P}_{u \leftrightarrow w}$ is implemented with multi-controlled NOT gates ($C^m(X)$). Using standard constructions, a $C^m(X)$ gate has a depth of $O(\log m) = O(\log \log |V|)$. The total circuit depth D_{circuit}

$$D_{\text{circuit}} = D \cdot \text{depth}(\mathcal{O}^{(j)}) = D \cdot O(\log \log |V|)$$

Gate count. The total gate count is the sum of gates over all D levels for all N_L paths. The QOTP operator $U(w_k)$ requires $O(n)$ single-qubit gates. A $C^m(X)$ gate can be decomposed into $O(m^2) = O((\log |V|)^2)$ elementary gates. The full operator thus requires $O((\log |V|)^2)$ gates. The total gate count $G_{\text{Broadcast}}$ is

$$\begin{aligned} G_{\text{Broadcast}} &= D \cdot N_L \cdot (\text{gates}(P_{\text{path}}) + \text{gates}(U_{\text{Enc}})) \\ &= D \cdot N_L \cdot (O((\log |V|)^2) + O(n)). \end{aligned}$$

VI. CONCLUSION

Our scheme exhibits favourable characteristics for limited-scale implementation on near-term quantum hardware. The total qubit requirement scales as $N_L(n + \lceil \log_2 |V| \rceil + 1)$, which remains tractable for small numbers of recipients and moderate message sizes, enabling proof-of-principle demonstrations on current NISQ platforms with approximately 50–200 physical qubits. The circuit depth is particularly well-suited for NISQ, growing only as $D \cdot O(\log \log |V|)$, and the dominant path transition gates decompose into manageable sequences of 2-local gates for small path register sizes. However, as the system scales to larger networks with increasing ,the cumulative gate count and multi-controlled gate overhead introduce significant noise sensitivity, quickly exceeding the coherent operation window of NISQ devices. In contrast, under near-fault-tolerant regimes with modest logical qubit overheads and active error correction, the scheme remains highly implementable even for substantially larger network sizes, benefiting from its inherently low logical circuit depth and parallelizable structure. Thus, while small-scale NISQ realizations are feasible for experimental validation, scalable deployment of the full protocol will require access to early-stage fault-tolerant quantum hardware. In the future, we aim to shrink the quantum resource footprint by lowering Hilbert-space dimensions and gate complexity as the number of messages grows, and to integrate fault-tolerant encodings to counteract noise in real-world quantum networks.

REFERENCES

- [1] A. Broadbent, C. E. González-Guillén, C. Schuknecht, A. Cabello, F. Dupuis, P. J. Lo, E. Kashefi, and I. Villanueva, “Quantum private broadcasting,” *Physical Review A*, vol. 105, no. 2, p. 022606, 2022.
- [2] A. Fiat and M. Naor, “Broadcast encryption,” in *Advances in Cryptology—CRYPTO’93: 13th Annual International Cryptology Conference Santa Barbara, California, USA August 22–26, 1993 Proceedings 13*. Springer, 1994, pp. 480–491.
- [3] J. Horwitz, “A survey of broadcast encryption,” *J. ACM*, 2003.
- [4] A. Sahai and B. Waters, “Fuzzy identity-based encryption,” in *Advances in Cryptology - EUROCRYPT 2005*, ser. LNCS, vol. 3494. Springer, 2005, pp. 457–473.
- [5] J. Bethencourt, A. Sahai, and B. Waters, “Ciphertext-policy attribute-based encryption,” in *IEEE Symposium on Security and Privacy (SP 2007)*. IEEE, 2007, pp. 321–334.
- [6] A. Lewko and B. Waters, “Decentralizing attribute-based encryption,” in *Annual international conference on the theory and applications of cryptographic techniques*. Springer, 2011, pp. 568–588.
- [7] V. Srivastava, S. K. Debnath, P. Stanica, and S. K. Pal, “A multivariate identity-based broadcast encryption with applications to the internet of things,” *Adv. Math. Commun.*, vol. 17, no. 6, pp. 1302–1313, 2023.
- [8] A. K. Singh, K. Acharya, and S. Mukhopadhyay, “Post-quantum secure recipient revocable broadcast encryption supporting anonymity,” *Multi-media Tools and Applications*, vol. 83, no. 2, pp. 4519–4531, 2024.
- [9] R. Sarkar, M. Mandal, and S. Mukhopadhyay, “Quantum-safe identity-based broadcast encryption with provable security from multivariate cryptography,” *Advances in Mathematics of Communications*, 2022.
- [10] H. Wee, “Optimal broadcast encryption and cp-abe from evasive lattice assumptions,” in *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. Springer, 2022, pp. 217–241.
- [11] W. Jian, Z. Quan, and T. Chao-Jing, “Quantum broadcast communication,” *Chinese Physics*, vol. 16, no. 7, p. 1868, 2007.
- [12] H. Qi, K. Sharma, and M. M. Wilde, “Entanglement-assisted private communication over quantum broadcast channels,” *Journal of Physics A: Mathematical and Theoretical*, vol. 51, no. 37, p. 374001, 2018.
- [13] Y. Yu, X. W. Zha, and W. Li, “Quantum broadcast scheme and multi-output quantum teleportation via four-qubit cluster state,” *Quantum Information Processing*, vol. 16, pp. 1–9, 2017.
- [14] N. Zhao, W. Li, and Y. Yu, “Quantum broadcast and multicast schemes based on partially entangled channel,” *IEEE Access*, vol. 8, pp. 29 658–29 666, 2020.
- [15] H. Sueno, T.-C. Wei, M. Hillery, J. A. Bergou, D. Fields, and V. S. Malinovsky, “Broadcasting single-qubit and multiqubit entangled states: Authentication, cryptography, and distributed quantum computation,” *Physical Review A*, vol. 107, no. 6, p. 062605, 2023.
- [16] W. Zhang, D. Qiu, X. Zou, and P. Mateus, “Analyses and improvement of a broadcasting multiple blind signature scheme based on quantum ghz entanglement,” *Quantum Information Processing*, vol. 16, pp. 1–23, 2017.
- [17] W. Zhang, D. Qiu, and X. Zou, “Improvement of a quantum broadcasting multiple blind signature scheme based on quantum teleportation,” *Quantum Information Processing*, vol. 15, pp. 2499–2519, 2016.
- [18] F. Salek, M.-H. Hsieh, and J. R. Fonollosa, “Single-serving quantum broadcast channel with common, individualized, and confidential messages,” *IEEE Transactions on Information Theory*, vol. 66, no. 12, pp. 7752–7771, 2020.
- [19] A. Broadbent *et al.*, “Quantum private broadcasting and unitary t-designs,” 2021. [Online]. Available: <https://arxiv.org/pdf/2107.11474>
- [20] C. Schuknecht, “Quantum private broadcasting,” Ph.D. dissertation, Université d’Ottawa/University of Ottawa, 2021.
- [21] A. Kaznatcheev, “Structure of exact and approximate unitary t-designs,” 2010.
- [22] C. Lancien and C. Majenz, “Weak approximate unitary designs and applications to quantum encryption,” *Quantum*, vol. 4, p. 313, 2020.
- [23] S. Samadder Chaudhury, A. Samanta, and A. Maitra, “Quantum attribute-based encryption: a comprehensive study,” *Quantum Information Processing*, vol. 22, no. 9, p. 335, 2023.
- [24] D. Aharonov, A. Ambainis, J. Kempe, and U. Vazirani, “Quantum walks on graphs,” in *Proceedings of the thirty-third annual ACM symposium on Theory of computing*, 2001, pp. 50–59.
- [25] C. Godsil and H. Zhan, “Discrete quantum walks,” Unpublished manuscript, 2020, <https://www.math.uwaterloo.ca/~cgodsil/quagmire/QuantumWalks/pdfs/DQW.pdf>.

- [26] A. M. Childs, D. Gosset, and Z. Webb, “Universal computation by multiparticle quantum walk,” *Science*, vol. 339, no. 6121, pp. 791–794, 2013.
- [27] A. Çakan, V. Goyal, C.-D. Liu-Zhang, and J. Ribeiro, “Computational quantum secret sharing,” *arXiv preprint arXiv:2305.00356*, 2023.
- [28] A. Ambainis, M. Mosca, A. Tapp, and R. De Wolf, “Private quantum channels,” in *Proceedings 41st Annual Symposium on Foundations of Computer Science*. IEEE, 2000, pp. 547–553.
- [29] Z. Liu, Z. Cao, and D. S. Wong, “Efficient generation of linear secret sharing scheme matrices from threshold access trees,” *Cryptology ePrint Archive*, 2010.
- [30] A. Shamir, “How to share a secret,” *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.

Algorithm 1 Generalized Policy-Based Quantum Broadcast Encryption and Decryption

- 1: **SETUP**($1^\lambda, \mathcal{A}_{\text{spec}}$):
 - 2: **Input:** Security parameter 1^λ , attributes, and policy specifications $\mathcal{A}_{\text{spec}}$.
 - 3: $(\text{PP}_{\text{PBE}}, \text{MSK}) \leftarrow \text{Setup}(1^\lambda, \mathcal{A}_{\text{spec}})$.
 - 4: Define tree $T = (V, E)$ with root r , N_L leaves $L = \{l_k\}$. For $k \in [1, N_L]$, unique path $P_k : v_0^{(k)} = r \rightarrow \dots \rightarrow v_{d_k}^{(k)} = l_k$. Max depth D .
 - 5: For $j \in [0, D]$, $v_j^{(k)}$ is the j -th node on P_k and note that $v_j^{(k)} = l_k$ if $j > d_k$.
 - 6: Hilbert spaces: $\mathcal{H}_V$ -path register, basis $\{|v\rangle\}_{v \in V}$, $\mathcal{H}_M$ -message space.
 - 7: Path operator: $P_{u \leftrightarrow w} := \begin{cases} |w\rangle\langle u| + |u\rangle\langle w| + \sum_{v' \in V \setminus \{u, w\}} |v'\rangle\langle v'| & \text{if } u \neq w \\ \mathbb{I}_V & \text{if } u = w \end{cases}$ on $\mathcal{H}_V$.
 - 8: Message encryption operators: Unitary $U(w)$ on $\mathcal{H}_M$ for target node w . If a path step is $l_k \rightarrow l_k$ i.e., path already at its leaf, the corresponding $U(l_k) = \mathbb{I}_M$.
 - 9: $\text{PP}_{\text{PQBE}} = (T, \{P_k\}, D, \mathcal{H}_V, \mathcal{H}_M, \{P_{u \leftrightarrow w}\}, \{U(w)\}, \text{PP}_{\text{ABE}})$.
 - 10: **KEY GENERATION**($\text{MSK}, \text{Policy}_{\text{user}}$): $\text{SK}_{\text{user}} \leftarrow \text{KeyGen}(\text{MSK}, \text{Policy}_{\text{user}})$.
 - 11: **ENCRYPTION**($\text{PP}_{\text{PQBE}}, \{\rho_k\}_{k=1}^{N_L}$):
 - 12: **Input:** PP_{PQBE} , messages $\{\rho_k \in \mathcal{D}(\mathcal{H}_M)\}_{k=1}^{N_L}$.
 - 13: Initial state: $\rho_B^{(0)} \leftarrow \bigotimes_{k=1}^{N_L} (|r\rangle\langle r|_{\mathcal{H}_V} \otimes \rho_k)$.
 - 14: **for** level $j = 0$ to $D - 1$ **do**
 - 15: For each path P_k , let $u_k = v_j^{(k)}, w_k = v_{j+1}^{(k)}$. The operators are $P_{op}^{(k)} = P_{u_k \leftrightarrow w_k}^{P_k}$ and $U_{op}^{(k)} = U(w_k)$. Note that these are $\mathbb{I}_V$ and $\mathbb{I}_M$ respectively if $j \geq d_k$.
 - 16: Global operator for level j : $\mathcal{O}^{(j)} \leftarrow \bigotimes_{k=1}^{N_L} (P_{op}^{(k)} \otimes U_{op}^{(k)})$.
 - 17: Update state: $\rho_B^{(j+1)} \leftarrow \mathcal{O}^{(j)} \rho_B^{(j)} (\mathcal{O}^{(j)})^\dagger$.
 - 18: **end for**
 - 19: Ciphertext: $\rho_{\text{Broadcast}} := \bigotimes_{k=1}^{N_L} (|l_k\rangle\langle l_k|_{\mathcal{H}_V} \otimes \rho_k') \leftarrow \rho_B^{(D)}$.
 - 20: Share all the keys along each path to users using the Shamir secret-sharing scheme.
 - 21: **DECRYPTION**($\text{PP}_{\text{PQBE}}, \text{SK}_{\text{user}}, \rho_{\text{Broadcast}}$):
 - 22: **Input:** $\text{PP}_{\text{PQBE}}, \text{SK}_{\text{user}}, \rho_{\text{Broadcast}}$.
 - 23: Determine set of authorized path indices I_{user} via SK_{user} . If $I_{\text{user}} = \emptyset$, output $\perp$.
 - 24: **for** each authorized index $m \in I_{\text{user}}$ **do**
 - 25: Extract m -th component: $\sigma_m^{(D)} = |l_m\rangle\langle l_m|_{\mathcal{H}_V} \otimes \rho_m' \leftarrow \text{Tr}_{\text{paths } k \neq m} [\rho_{\text{Broadcast}}]$.
 - 26: Let $P_m : v_0^{(m)} = r \rightarrow \dots \rightarrow v_{d_m}^{(m)} = l_m$ be the m -th path.
 - 27: The user reconstructs the keys associated with a particular path using the Shamir secret-sharing scheme.
 - 28: **for** level $j = D - 1$ down to 0 **do**
 - 29: Define $u_m = v_j^{(m)}, w_m = v_{j+1}^{(m)}$.
 - 30: Operators for that forward step: $P_{op} = P_{u_m \leftrightarrow w_m}$ and $U_{op} = U(w_m)$.
 - 31: Reverse transformation: $\sigma_m^{(j)} \leftarrow (P_{op}^\dagger \otimes U_{op}^\dagger) \sigma_m^{(j+1)} (P_{op} \otimes U_{op})$.
 - 32: **end for**
 - 33: $\sigma_m^{(0)} = |r\rangle\langle r|_{\mathcal{H}_V} \otimes \rho_m$. Original message: $\rho_m \leftarrow \text{Tr}_{\mathcal{H}_V} [\sigma_m^{(0)}]$.
 - 34: **end for**
 - 35: Output all successfully decrypted messages $\{\rho_m\}_{m \in I_{\text{user}}}$.
-